TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Standardization of Sodium Thiosulphate with Potassium Dichromate

Why thiosulphate must be certified before it can measure anything, and the indirect iodometric chain — dichromate to iodine to thiosulphate — that certifies it.

CHE-1206 · CHEMISTRY-II SESSIONAL · EXPERIMENT 5 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to determine the exact strength of an approximately 0.1 N sodium thiosulphate solution by titrating it against iodine liberated from a standard potassium dichromate solution.

Sodium thiosulphate cannot be its own standard. The pentahydrate, $Na_2S_2O_3 \cdot 5H_2O$, effloresces and its crystals are not of guaranteed composition, and the made-up solution is attacked by bacteria, dissolved CO_2, light and air — its strength drifts from the day it is prepared. Potassium dichromate, by contrast, is an excellent **primary standard**: pure, stable, non-hygroscopic, and weighable to an exact solution. The standardisation is *indirect* (iodometric): dichromate does not react conveniently with thiosulphate directly, so in acid solution it is made to oxidise iodide, releasing an exactly equivalent amount of iodine —

$$Cr_2O_7^{2-} + 6\ I^- + 14\ H^+ \rightarrow 2\ Cr^{3+} + 3\ I_2 + 7\ H_2O$$

— and the liberated iodine is titrated with the thiosulphate, with starch as indicator:

$$I_2 + 2\ S_2O_3^{2-} \rightarrow 2\ I^- + S_4O_6^{2-}$$

Chaining the two equations: **1 mol $Cr_2O_7^{2-}$ ≡ 3 mol I_2 ≡ 6 mol $S_2O_3^{2-}$**. Dichromate takes 6 electrons per formula unit, so its equivalent weight is 294.18/6 = **49.03 g**, and a 0.1000 N solution contains 4.903 g L^{-1}. Thiosulphate loses one electron per ion ($2\ S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2\ e^-$), so for thiosulphate normality and molarity are the same number. Then N_1V_1 (dichromate) = N_2V_2 (thiosulphate) closes the calculation, as in Chemistry-I.

Why the conditions matter. The dichromate–iodide reaction is slow, so the acidified mixture stands stoppered **in the dark for about 5 minutes**: time to finish, and darkness because light and air oxidise iodide in acid solution and would invent extra iodine. The mixture is then **diluted** before titrating — in

strong acid, thiosulphate itself decomposes ($S_2O_3^{2-} + 2\ H^+ \rightarrow S + SO_2 + H_2O$) and air-oxidation of iodide accelerates. Starch is held back until the brown iodine colour has faded to pale straw yellow, for reasons detailed in the precautions.

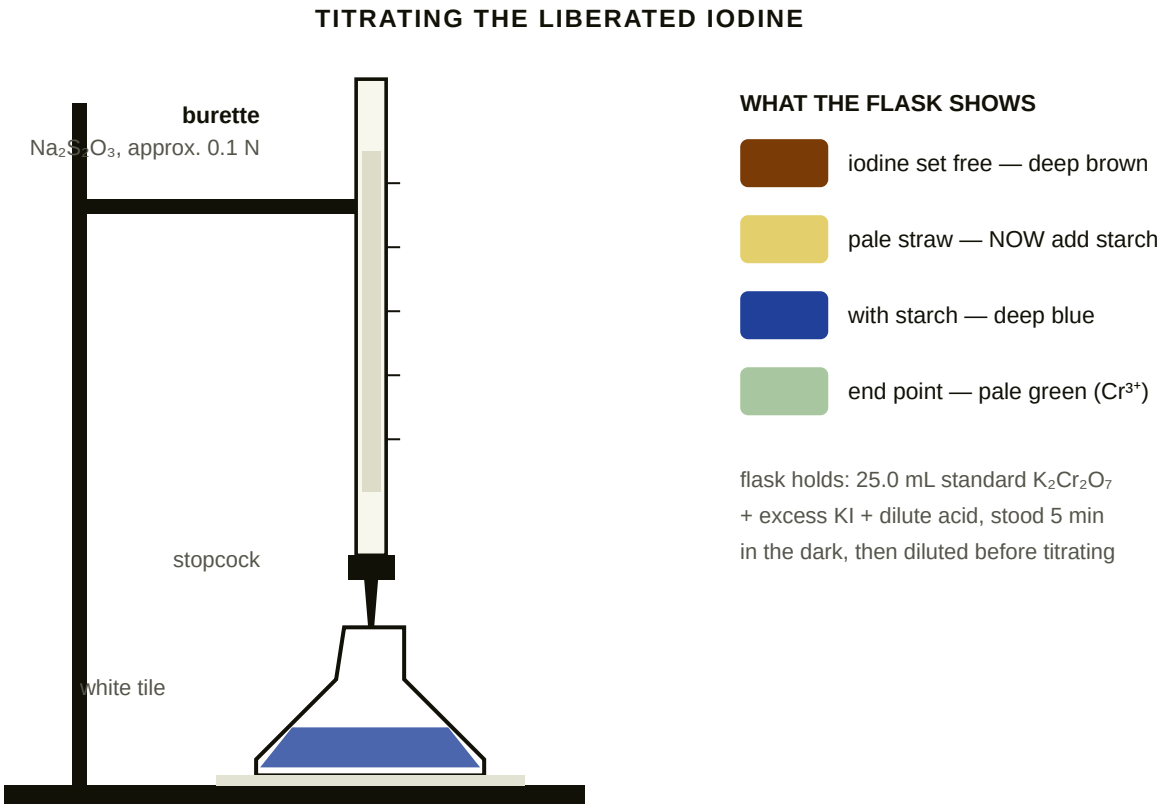


Figure 1 — The iodometric titration. The end point is not colourless: the blue of the starch–iodine complex vanishes into the pale green of the Cr^{3+} formed in the first reaction.

2. Apparatus and solutions

- Burette, 25 mL pipette, conical flasks with stoppers (or watch glasses), measuring cylinders, white tile.
- **Standard $K_2Cr_2O_7$, 0.1000 N:** 4.903 g of the dry salt per litre (equivalent weight 49.03).
- **Sodium thiosulphate, approx. 0.1 N:** about 24.8 g of $Na_2S_2O_3 \cdot 5H_2O$ per litre, made with freshly boiled and cooled distilled water; this is the solution under standardisation and it fills the burette.
- Potassium iodide solution, 10% (must be colourless — yellowed KI already contains iodine); dilute sulphuric acid (about 2 M) or dilute hydrochloric acid; starch solution, 1%, freshly prepared.

3. Procedure

1. Rinse and fill the burette with the thiosulphate solution; remove the funnel; check the jet for air bubbles.

2. Pipette 25.0 mL of the standard dichromate into a conical flask. Add 10 mL of 10% KI solution and 10 mL of dilute acid. Stopper the flask, swirl, and leave it **in the dark for about 5 minutes** — the solution is deep brown with liberated iodine.
3. Dilute with about 100 mL of distilled water. Titrate at once with the thiosulphate, swirling steadily, until the brown fades to a **pale straw yellow**.
4. Add 1–2 mL of starch solution: the mixture turns **deep blue**. Continue the titration drop by drop until the blue just disappears and the solution is **pale green** (the colour of Cr^{3+}). Record the titre.
5. Repeat until two titres agree within 0.1 mL; average the concordant pair.

4. Worked calculation (model data)

STANDARDISATION RESULT

Standard $K_2Cr_2O_7$: 0.1000 N. Volume pipetted: 25.0 mL. Concordant titre of $Na_2S_2O_3$: 24.8 mL.
Equivalents must match: $N(K_2Cr_2O_7) \times V(K_2Cr_2O_7) = N(Na_2S_2O_3) \times V(Na_2S_2O_3)$.
$0.1000 \times 25.0 = N \times 24.8 \rightarrow N = 2.500/24.8 =$ **0.1008 N**.
Chain check by moles: 25.0 mL of 0.1000 N dichromate = 2.500×10^{-3} equivalents = 4.167×10^{-4} mol $Cr_2O_7^{2-}$, which frees 1.250×10^{-3} mol I_2, which consumes 2.500×10^{-3} mol $S_2O_3^{2-}$ — in 24.8 mL, that is $2.500 \times 10^{-3}/0.0248 =$ **0.1008 M**, the same figure, as it must be (1 mol thiosulphate = 1 equivalent).
Label the bottle "$Na_2S_2O_3$, 0.1008 N, standardised (date)" — Experiment 6 spends this number.

5. Observation table (as recorded)

TRIAL	INITIAL READING (ML)	FINAL READING (ML)	TITRE (ML)
1 (rough)	0.0	25.2	25.2
2	0.0	24.8	24.8
3	0.0	24.8	24.8

Concordant titre: 24.8 mL. Note how the standardisation borrows the dichromate's primary-standard certainty through the iodine: the thiosulphate never reacts with dichromate directly, yet its strength is anchored to a weighed salt.

6. Precautions

- **Starch goes in late — only when the solution is pale straw yellow.** Added early, the starch binds a large amount of iodine into a complex from which the iodine is released only slowly; the end point then drags and is overshot. In strong acid, starch also hydrolyses and loses sensitivity.
- Keep the acidified iodide mixture **stoppered and in the dark** for the standing time: acid + air + light oxidise I^- to I_2 ($4\ I^- + O_2 + 4\ H^+ \rightarrow 2\ I_2 + 2\ H_2O$), inventing iodine that no dichromate authorised — the thiosulphate then reads stronger than it is.
- **Dilute before titrating.** Strongly acid solution decomposes the titrant itself ($S_2O_3^{2-} + 2\ H^+ \rightarrow S\downarrow + SO_2 + H_2O$) and speeds the air-oxidation above; dilution also thins the Cr^{3+} green so the end point is easier to judge.
- Titrate promptly once diluted: iodine is volatile and escapes from an open, standing flask.
- Use the acid quantity given — no more. Great excess acid worsens every side reaction named above without helping the main one.
- Read the burette at the **top** of the meniscus if the dark solution ever hides the bottom; here the titrant is colourless, so read the bottom of the meniscus in the usual way, at eye level, funnel removed.

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is sodium thiosulphate not a primary standard, while potassium dichromate is?
2. Why does the acidified dichromate–iodide mixture stand in the dark for five minutes before titration?
3. Why is starch withheld until the solution has faded to pale straw yellow?
4. Why is the mixture diluted before the titration begins?
5. 20.0 mL of 0.1000 N $K_2Cr_2O_7$ liberates iodine requiring 19.6 mL of a thiosulphate solution. Find the normality of the thiosulphate.
6. The end point here is pale green, not colourless. Where does the green come from, and what would a returning blue colour after the end point mean?

Answer key

1. The pentahydrate effloresces and is of uncertain composition, and its solutions are decomposed by bacteria, dissolved CO_2, light and air, so neither a weighing nor an old bottle can be trusted. $K_2Cr_2O_7$ is obtained pure, is stable in air and in solution, and does not absorb water — a weighed amount is an exactly known amount.
2. The reaction $Cr_2O_7^{2-} + 6\ I^- + 14\ H^+ \rightarrow 2\ Cr^{3+} + 3\ I_2 + 7\ H_2O$ is comparatively slow, so time is allowed for it to complete. Darkness (and the stopper) prevents the competing air-and-light oxidation of iodide in acid, which would liberate extra iodine and corrupt the equivalence.

3. The starch–iodine complex is very strong. With much iodine present, iodine is held in the complex and released too slowly near the end point, giving a dragging, overshot finish; strong acid also hydrolyses starch. When only a little iodine remains (pale yellow), the small amount complexed releases cleanly and the blue vanishes sharply.
4. In strong acid the titrant decomposes ($S_2O_3^{2-} + 2\ H^+ \rightarrow S + SO_2 + H_2O$) and iodide is air-oxidised faster; dilution suppresses both, and it also lightens the Cr^{3+} colour so the disappearance of blue is visible.
5. $N \times 19.6 = 0.1000 \times 20.0 \rightarrow N = 2.000/19.6 =$ **0.1020 N**.
6. The green is Cr^{3+}, the reduction product of the dichromate, present from the first reaction and unmasked when the blue starch–iodine colour is discharged. A blue colour creeping back on standing means air has oxidised iodide in the still-acid solution to fresh iodine — the titration was finished; the return is an artefact, not a missed end point.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib